BSc 2023/2024

For internal students of Imperial College.

Mathematics Foundations (BUSI40001)- Resit Exam

2 hours

XXXday ABth September 2024, 09:00 – 11:00

Instructions for the exam

- For Section A, internet access with ChatGPT is allowed and a computer with R or Python is needed.
- Section B is closed-book and paper-based. Section B is to be made accessible to the students after completing Section A.
- Answer ALL questions (Please note that not all questions carry equal marks, the mark breakdown is indicated throughout the exam).
- It is important to write neatly and clearly.
- If a question or part of a question is unclear to you, state any assumptions you need in order to complete the question.

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SECTION A

You have 45 minutes for Section A. Answer all questions. You must close your laptop after 45 minutes.

1. (19 marks) The maths and physics marks, of 7 students are shown in the table below.

Student	A	B	C	D	E	F	G
Maths	100	85	78	35	80	45	24
Physics	92	73	65	27	89	40	21

This question is about estimating a physics mark given a maths mark.

(a) Write out the linear regression in the matrix form

$$\underline{y} = \mathbf{X}\underline{\beta} + \underline{\epsilon},$$

where $\underline{y}$, $\mathbf{X}$, $\underline{\beta}$, and $\underline{\epsilon}$ are to be defined. [2 marks]

[1/2 a mark for each vector]

$$\underline{y} = (92, 73, 65, 27, 89, 40, 21)^\top$$

is the 7×1 vector of physics marks,

$$\underline{x} = (100, 85, 78, 35, 80, 45, 24)^\top$$

is the 7×1 vector of maths marks, $\underline{e} = (1, \ldots, 1)^\top$ is 7×1 vector of ones.

$$\mathbf{X} = [\underline{e}, \underline{x}]$$

is the 7 by 2 design matrix,

$$\underline{\beta} = [\beta_1, \beta_2]^\top$$

is the 1 by 2 parameter vector, and $\underline{\epsilon}$ is 7×1 error vector.

(b) Show that the expression for $\underline{\beta}$ which minimizes $\underline{\epsilon}^\top \underline{\epsilon}$ is given by

$$\widehat{\underline{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\underline{y}.$$

[5 marks]

Observe that $\underline{\epsilon}^\top \underline{\epsilon}$, where $\underline{\epsilon} = \underline{y} - \mathbf{X}\underline{\beta}$, is minimized by projecting $\underline{y}$ orthogonally into the column space of $\mathbf{X}$. Therefore, we want to choose $\underline{\beta}$ such that $\mathbf{X}^\top(\underline{y} - \mathbf{X}\underline{\beta}) = \underline{0}$. We shall denote the parameter vector which satisfies the latter equation via $\widehat{\underline{\beta}}$ and so

$$\mathbf{X}^\top(\underline{y} - \mathbf{X}\widehat{\underline{\beta}}) = \underline{0}$$

Therefore,

$$\mathbf{X}^\top\underline{y} = \mathbf{X}^\top\mathbf{X}\widehat{\underline{\beta}}.$$

Now $\mathbf{X}^\top\mathbf{X}$ is a 2 by 2 square matrix. Now

$$\widehat{\underline{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y}.$$

(c) Show that the fitted values are given by

$$\underline{\hat{y}} = \mathbf{H}\underline{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$$

[3 marks]

The fitted values of the dependent variable are given by $\underline{\hat{y}} = \underline{\widehat{\beta}}\mathbf{X}$. Therefore,

$$\underline{\hat{y}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y},$$

and so

$$\underline{\hat{y}} = \mathbf{H}\underline{y},$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T.$$

(d) Show that $\mathbf{H}$ is idempotent and symmetric. [4 marks]

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is idempotent. Now,

$$\begin{aligned}
\mathbf{H}^\top &= [\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T]^\top \\
&= [\mathbf{X}^\top]^\top[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top\mathbf{X}^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^\top]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top(\mathbf{X}^T)^\top\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is symmetric.

(e) Estimate the physics mark of a student who scored 98 in the maths test, but was absent from the physics test. You should do this using either R or Python [5 marks]

You need a calculator or a computer!

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 7 & 447 \\ 447 & 33535 \end{pmatrix}$$

$$\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1} = \begin{pmatrix} \frac{33535}{34936} & -\frac{447}{34936} \\ -\frac{447}{34936} & \frac{7}{34936} \end{pmatrix}$$

$$\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top = \begin{pmatrix} -\frac{1015}{3176} & -\frac{1115}{8734} & -\frac{121}{3176} & \frac{8945}{17468} & -\frac{2225}{34936} & \frac{305}{794} & \frac{22807}{34936} \\ \frac{23}{3176} & \frac{37}{8734} & \frac{9}{3176} & -\frac{101}{17468} & \frac{113}{34936} & -\frac{3}{794} & -\frac{279}{34936} \end{pmatrix}$$

$$\begin{aligned} \underline{\widehat{\beta}} &= \left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top\underline{y} \\ &= \left(-\frac{12593}{3176}, \frac{3089}{3176}\right)^\top \end{aligned}$$

$$\beta_1 + \beta_2 98 = 91$$

Estimated physics mark is 91.

2. (20 marks) Find and classify all the critical points of the function $f(x, y) = x^2 - y^2$ and determine their type, where $x, y \in \mathbb{R}$. Use Python or R to draw a plot of the surface and check your written calculations.

We note that

$$f(x + \Delta x, x + \Delta y) = f(x, y) + \underline{\nabla} f(x, y)\underline{\Delta x} + \underline{\Delta x}^\top \mathbf{H}(x, y)\underline{\Delta x} + \ldots,$$

where $\underline{\Delta x} = (\Delta x, \Delta y)^\top$. To find local optima, we solve $\underline{\nabla} f(x, y) = (0, 0)$, i.e.

$$\begin{aligned} 2x &= 0 \\ -2y &= 0. \end{aligned}$$

We see that $(0, 0)$ is a local optimum. Now we find $\mathbf{H}(0, 0)$ and its eigenvalues.

$$\mathbf{H}(0, 0) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

Therefore the eigenvalues are clearly -2 and 2, and so $(0, 0)$ is a saddlepoint.

The following Python code finds local optima and classifies them.

```
import sympy as sp
import matplotlib.pyplot as plt
import numpy as np
from mpl_toolkits.mplot3d import Axes3D

# Define the symbols
x, y = sp.symbols('x y')

# Define the function
f = x**2 - y**2

# Compute the first derivatives (gradient)
grad_f = [sp.diff(f, var) for var in (x, y)]

# Solve for critical points
critical_points = sp.solve(grad_f, (x, y))

# Ensure critical points are in the correct format
if isinstance(critical_points, dict):
    # Convert dictionary to list of tuples
    critical_points = [tuple(critical_points.values())]
elif isinstance(critical_points, tuple):
    # Enclose tuple in a list
    critical_points = [critical_points]

# Compute the Hessian matrix
H = sp.hessian(f, (x, y))

# Function to determine the nature of critical points
```

```python
def nature_of_critical_point(Hessian, point):
    # Substitute the point into the Hessian
    H_sub = Hessian.subs({x: point[0], y: point[1]})

    # Compute eigenvalues
    eigenvals = H_sub.eigenvals()

    # Check if all eigenvalues are positive, negative, or mixed
    if all(val > 0 for val in eigenvals):
        return "Local Minimum"
    elif all(val < 0 for val in eigenvals):
        return "Local Maximum"
    else:
        return "Saddle Point"

# Analyze each critical point
results = [(point, nature_of_critical_point(H, point)) for point in critical_points]

# Define the function for plotting
def f_plot(x, y):
    return x**2 - y**2

# Create a grid of x and y values for plotting
x_vals = np.linspace(-3, 3, 400)
y_vals = np.linspace(-3, 3, 400)
x_vals, y_vals = np.meshgrid(x_vals, y_vals)

# Calculate z values for the grid
z_vals = f_plot(x_vals, y_vals)

# Create plots
fig = plt.figure(figsize=(12, 6))

# 3D surface plot
ax1 = fig.add_subplot(121, projection='3d')
ax1.plot_surface(x_vals, y_vals, z_vals, cmap='viridis', edgecolor='none')
ax1.set_title('3D Surface Plot')
ax1.set_xlabel('X axis')
ax1.set_ylabel('Y axis')
ax1.set_zlabel('Z axis')

# Contour plot
ax2 = fig.add_subplot(122)
contour = ax2.contourf(x_vals, y_vals, z_vals, cmap='viridis')
ax2.plot(0, 0, 'ro') # mark the saddle point
ax2.set_title('Contour Plot with Saddle Point')
ax2.set_xlabel('X axis')
ax2.set_ylabel('Y axis')

# Add a color bar for the contour plot
fig.colorbar(contour, ax=ax2, orientation='vertical')

# Show the plot
plt.show()

results
```

SECTION B

You have 1 hour 15 minutes for Section B. Answer all questions.

1. (30 marks) Consider a real sequence $(a_n)_n$

(a) What does it mean to say that (a_n) converges to a limit l ? [2 marks]

Standard definition: all students should know this.
a_n converges to L if for all $\varepsilon > 0$, there is some $N \in \mathbb{N}$ such that whenever $n \geq N$, we have $|a_n - l| < \varepsilon$. In symbols, this says

$$(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - l| < \varepsilon.$$

(b) Consider a real sequence (a_n). For all $n \in \mathbb{N}$, define $b_n = \frac{\sum_{i=1}^{n} a_i}{n}$ and thereby create the new sequence (b_n).
Prove that if $\lim a_n = l$, then $\lim b_n = l$ [12 marks]

We need to work with $|b_n - l|$

$$|b_n - l| = \left| \frac{a_1 + \cdots + a_n}{n} - l \right| = \frac{|a_1 - l + \cdots + a_n - l|}{n}$$

Using the triangle inequality repeatedly

$$|a_1 - l + \cdots + a_n - l| \leq \sum_{i=1}^{n} |a_i - l|$$

Now $\lim a_n = l$, and so

$$(\forall \epsilon > 0)(\exists N_1 \in \mathbb{N})(\forall n \geq N_1)|a_n - l| < \frac{\epsilon}{2}.$$

Therefore, for all $n \geq N_1$

$$|a_1 - l + \cdots + a_n - l| < \sum_{i=1}^{N_1} |a_i - l| + (n - N_1)\frac{\epsilon}{2}.$$

Now define

$$D = \sup \{|a_n - l| : n \in \mathbb{N}\}.$$

Then,

$$(n \geq N_1)|b_n - l| < \frac{DN_1 + (n - N_1)\frac{\epsilon}{2}}{n} = \frac{N_1}{n} D + \left(1 - \frac{N_1}{n}\right)\frac{\epsilon}{2},$$

where the RHS is between D and $\frac{\epsilon}{2}$.
For all $\epsilon > 0$, we have either $D > \epsilon$ or $D \leq \epsilon$. When $D > \epsilon$, we can find an N_2 such that $\frac{N_1}{n} D + \left(1 - \frac{N_1}{n}\right)\frac{\epsilon}{2} < \epsilon$ and for all $n \geq N_2$, $\frac{N_1}{n} D + \left(1 - \frac{N_1}{n}\right)\frac{\epsilon}{2} < \epsilon$. Therefore, by picking $N = \max\{N_1, N_2\}$

$$(\forall \epsilon \in (0, D))(\exists N \in \mathbb{N})(\forall n \geq N)|b_n - l| < \epsilon$$

N.B $\epsilon \in (0, D)$ just means $0 < \epsilon < D$.
When $D \leq \epsilon$ it is easy to see $\frac{N_1}{n} D + \left(1 - \frac{N_1}{n}\right)\frac{\epsilon}{2} < \epsilon$ and so

$$(\forall \epsilon \geq D)(\exists N \in \mathbb{N})(\forall n \geq N)|b_n - l| < \epsilon$$

Hence,

$$(\forall \epsilon \geq 0)(\exists N \in \mathbb{N})(\forall n \geq N)|b_n - l| < \epsilon.$$

(c) Define what is meant by a subsequence of $(a_n)_n$. [3 marks]

Let (a_n) be a sequence of real numbers, and let $n_1 < n_2 < n_3 < n_4 < n_5 < \ldots$ be a strictly increasing sequence of natural numbers. Then the sequence

$$(a_{n_1}, a_{n_2}, a_{n_3}, a_{n_4}, a_{n_5}, \ldots)$$

is called a subsequence of (a_n) and is denoted by (a_{n_k}), where $k \in \mathbb{N}$ indexes the subsequence.

(d) Give an example of a sequence $(b_n)_{n\geqslant 1}$ and a subsequence $(b_{n_k})_{k\geqslant 1}$ such that $(b_{n_k})_{k\geqslant 1}$ converges to a limit l_1 and the sequence $(b_{2n_k})_{k\geqslant 1}$ converges to a limit $l_2 \neq l_1$. [4 marks]

For $n \in \mathbb{Z}^+$, define $b_n = (-)^n$. The subsequence b_{n_k} where $n_k = 2k+1$ is given by $b_{n_k} = b_{2k+1} = -1$, which clearly converges to -1. The sequence b_{2n_k} is given by $b_{2n_k} = 1$, which clearly converges to 1.

(e) What does it mean to say that (a_n) is bounded ? [2 marks]

Standard definition: all students should know this.
A sequence (a_n) is bounded if

$$(\exists L, U \in \mathbb{R})(\forall n \in \mathbb{N}) L \leq a_n \leq U$$

.

(f) Show that the sequence $(a_n)_{n\geqslant 1}$ defined by $a_n = \left(1 + \frac{1}{n}\right)^n$ is bounded. [7 marks]

$$(\forall n \in \mathbb{Z}^+)(a_{n+1} \geq a_n) \wedge a_1 = 2 \Rightarrow (\forall n \in \mathbb{Z}^+)(a_n \geq 2)$$

Using the Binomial Theorem, for all

$$(\forall n \in \{0, 1, 2, \ldots\}) \left(a_n = \sum_{r=0}^{n} \binom{n}{r} \frac{1}{n^r} \right)$$

$$(\forall n \in \mathbb{Z}^+) \left(\binom{n}{r} \frac{1}{n^r} = \frac{n(n-1)\ldots(n-r+1)}{r!} n^r \leq \frac{n^r}{r!} n^r = \frac{1}{r!} \right)$$

Therefore

$$(\forall n \in \mathbb{Z}^+) \left(a_n \leq \sum_{r=0}^{n} \frac{1}{r!} = 1 + \sum_{r=1}^{n} \frac{1}{r!} \leq 1 + \sum_{r=1}^{n} \frac{1}{2^{r-1}} \right)$$

Hence,

$$(\forall n \in \mathbb{Z}^+) \left(a_n \leq 1 + \frac{1 - \frac{1}{2^n}}{1 - \frac{1}{2}} = 1 + 2 - \frac{1}{2^{n-1}} \leq 3 \right)$$

Therefore,

$$(\forall n \in \mathbb{Z}^+)(2 \leq a_n \leq 3),$$

and so, by definition, $(a_n)_n$ is a bounded sequence.

2. (24 marks)

Let V be a vector space over $\mathbb{R}$ with inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$.

(a) Define what is meant by an inner product. [3 marks]

An **inner product** is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$, which satisfies the following 3 axioms $(\forall x, y, z \in V)(\forall a \in \mathbb{R})$:

i. $\langle x, x \rangle \geq 0$ with equality iff $x = 0$ (positive definiteness)
ii. $\langle ax, y \rangle = a\langle x, y \rangle$ (homogeneity in the 1st argument)
iii. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$ (additivity in the 1st argument)

(b) Define what is meant by a norm on a vector space V over $\mathbb{R}$. [3 marks]

A **norm** on a vector space V over $\mathbb{R}$ is a non - negative function $\|\cdot\| : V \to \mathbb{F}$ with the properties that:

i. $\|x\| = 0$ if and only if $x = 0$ (positive definite)
ii. $\forall x, y \in V, \|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)
iii. $\forall \lambda \in \mathbb{R} \& x \in V, \|\lambda x\| = |\lambda| \|x\|$

(c) Define $w = x - \lambda y$ where $\lambda \in \mathbb{R}$ and $x, y \in V$
Consider $f(\lambda) = \langle w, w \rangle \in \mathbb{R}$ as a function of λ.
Find the value of λ which maximizes $f(\lambda)$. Hence, prove that $|\langle x, y \rangle| \leq \sqrt{\langle x, x \rangle}\sqrt{\langle y, y \rangle}$. [10 marks]

(d) Show that $\|x\| = \sqrt{\langle x, x \rangle}$ is a norm. [4 marks]

$\langle x, x \rangle = 0$ iff $x = 0$ from postive definiteness property of an inner product.
Suppose $x, y \in V$.
$\langle x + y, x + y \rangle = \langle x, x \rangle + 2\langle x, y \rangle + \langle y, y \rangle \leq \langle x, x \rangle + 2\sqrt{\langle x, x \rangle \langle y, y \rangle} + \langle y, y \rangle = (\sqrt{\langle x, x \rangle} + \langle y, y \rangle)^2$
Therefore $\langle x + y, x + y \rangle \leq (\sqrt{\langle x, x \rangle} + \sqrt{\langle y, y \rangle})^2$ with equality iff $\langle x, y \rangle = 0$. Hence,

$$\|x + y\| \leq \|x\| + \|y\|.$$

This holds true for all $x, y \in V$. So we have established the triangle inequality.
Now

$$\langle \lambda x, \lambda x \rangle = \lambda^2 \langle x, x \rangle$$

and so
$\lambda \|x\| = |\lambda| \|x\|$.

(e) Suppose V is finite dimensional. Let $\mathcal{S}$ be a finite subset of V. What does it mean for $\mathcal{S}$ to be

i. a linearly independent set [2 marks]

The vectors in a set $\mathcal{S} = \{x_1, \ldots, x_n\}$ are said to be **linearly independent** if

$$\sum_{i=1}^{n} \lambda_i x_i = 0 \iff (\forall i \in \{1, \ldots, n\})\lambda_i = 0.$$

ii. a spanning set for V [2 marks]

A set of vectors $\mathcal{S} = \{v_1, \ldots, v_n\}$ in a vector space V over $\mathbb{R}$ is said to be a **spanning set** of V if every vector in V can be expressed as a linear combination of the vectors in $\mathcal{S}$. In other words, $\mathcal{S}$ spans V (or $\mathcal{S}$ is a spanning set of V)

$$(\forall v \in V)(\exists a_1, a_2, \ldots, a_n \in \mathbb{R})v = \sum_{i=1}^{n} a_i v_i.$$

End of Paper